P7: Trigonometric Identities and Equations

1. Fundamental Trigonometric Identities

Pythagorean Identity (Year 1 core):

$$\sin^2\theta + \cos^2\theta \equiv 1$$

This holds for all values of θ. The $\equiv$ symbol means "identically equal" — true for every value, not just specific solutions.

Dividing both sides of $\sin^2\theta + \cos^2\theta \equiv 1$ by $\cos^2\theta$ gives:

$$\tan^2\theta + 1 \equiv \sec^2\theta$$

Dividing instead by $\sin^2\theta$ gives:

$$1 + \cot^2\theta \equiv \csc^2\theta$$

Tangent Identity:

$$\tan\theta \equiv \frac{\sin\theta}{\cos\theta}$$

Undefined when $\cos\theta = 0$, i.e., at $\theta = 90^\circ, 270^\circ, \ldots$

Using Identities to Simplify and Prove

These identities allow you to rewrite expressions entirely in terms of one trig function, which is essential for solving equations and proving results.

Example — Prove that $\frac{\sin^2\theta}{1-\cos\theta} \equiv 1+\cos\theta$:

LHS $= \dfrac{\sin^2\theta}{1-\cos\theta}$

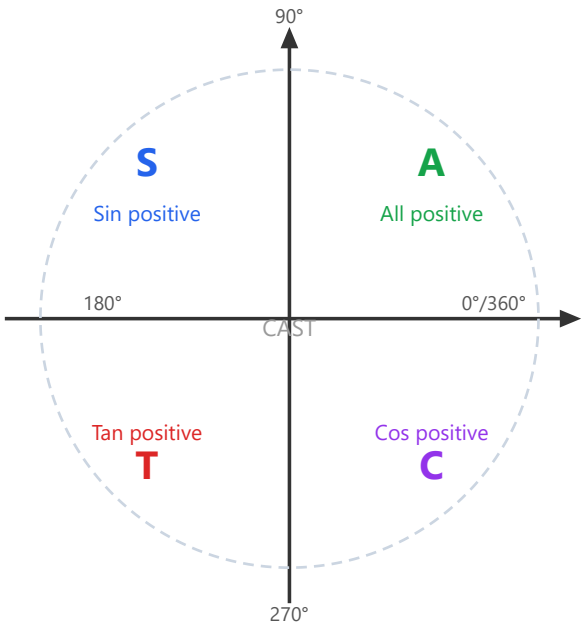
Substitute $\sin^2\theta = 1-\cos^2\theta = (1-\cos\theta)(1+\cos\theta)$:

$= \dfrac{(1-\cos\theta)(1+\cos\theta)}{1-\cos\theta} = 1+\cos\theta$ = **RHS** ✓

Exam Tip: Always start from the more complicated side in a proof. Never move terms across the equals sign — work on each side independently until they match.

2. The CAST Diagram

The CAST diagram shows which trig ratios are **positive** in each quadrant (angles measured anticlockwise from positive x-axis).



Memory aid: "All Students Take Chemistry" — going anticlockwise from quadrant 1.

Quadrant	Angle range	Positive ratios
1st (A)	$0^\circ < \theta < 90^\circ$	sin, cos, tan
2nd (S)	$90^\circ < \theta < 180^\circ$	sin only
3rd (T)	$180^\circ < \theta < 270^\circ$	tan only
4th (C)	$270^\circ < \theta < 360^\circ$	cos only

Finding All Solutions Using CAST

If the principal value (from calculator) is α, the second solution in $[0^\circ, 360^\circ]$ is:

- For $\sin\theta = k$: second solution is $180^\circ - \alpha$
- For $\cos\theta = k$: second solution is $360^\circ - \alpha$
- For $\tan\theta = k$: second solution is $180^\circ + \alpha$

Exam Tip: Always check how many solutions are expected in the given interval. For $0^\circ \leq x \leq 360^\circ$ there are usually 2 solutions for sin and cos equations, 2 for tan equations.

3. Solving Trigonometric Equations

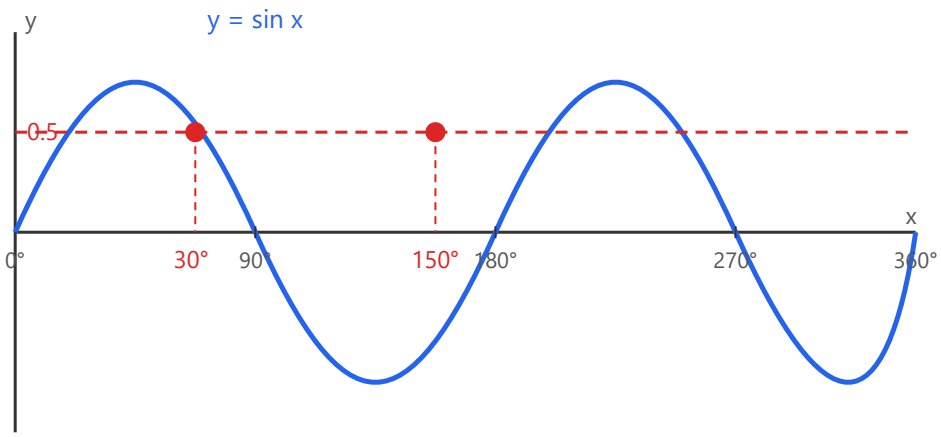
Basic Method

1. Rearrange to get a single trig function = number
2. Find the principal value using your calculator ($\sin^{-1}$, $\cos^{-1}$, $\tan^{-1}$)
3. Use CAST or graph to find all solutions in the given interval
4. Check all solutions satisfy the original equation

Example — Solve $\sin x = 0.5$ for $0^\circ \leq x \leq 360^\circ$:

Principal value: $x = \sin^{-1}(0.5) = 30^\circ$

Since sin is positive, solutions are in Q1 and Q2:
$x = 30^\circ$ and $x = 180^\circ - 30^\circ = 150^\circ$



Equations Involving Transformations

When solving $\sin(2x) = 0.5$ for $0^\circ \leq x \leq 360^\circ$, let $u = 2x$, so the interval for u becomes $0^\circ \leq u \leq 720^\circ$.

Example — Solve $\sin(2x) = 0.5$ for $0^\circ \leq x \leq 360^\circ$:

Let $u = 2x$, interval: $0^\circ \leq u \leq 720^\circ$
$\sin u = 0.5 \Rightarrow u = 30^\circ, 150^\circ, 390^\circ, 510^\circ$
Divide by 2: $x = 15^\circ, 75^\circ, 195^\circ, 255^\circ$

4. Disguised Quadratics in Trig

Many Edexcel exam questions disguise a quadratic equation using trig functions. Spot these by looking for squared trig terms.

Example — Solve $2\sin^2 x - \sin x - 1 = 0$ for $0^\circ \leq x \leq 360^\circ$:

Let $s = \sin x$. Then: $2s^2 - s - 1 = 0$
Factorise: $(2s + 1)(s - 1) = 0$
So $s = -\frac{1}{2}$ or $s = 1$

Case 1: $\sin x = 1 \Rightarrow x = 90^\circ$
Case 2: $\sin x = -\frac{1}{2}$: sin is negative, so solutions in Q3 and Q4:
$x = 180^\circ + 30^\circ = 210^\circ$ and $x = 360^\circ - 30^\circ = 330^\circ$

Solutions: $x = 90^\circ, 210^\circ, 330^\circ$

Example — Solve $2\cos^2\theta - 3\cos\theta + 1 = 0$ for $0 \leq \theta \leq 2\pi$:

Let $c = \cos\theta$: $(2c-1)(c-1) = 0$
$\cos\theta = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{3}, \frac{5\pi}{3}$
$\cos\theta = 1 \Rightarrow \theta = 0, 2\pi$

Solutions: $\theta = 0, \frac{\pi}{3}, \frac{5\pi}{3}, 2\pi$

Exam Tip: When the equation involves both $\sin^2 x$ and $\cos^2 x$, use the identity $\sin^2 x = 1 - \cos^2 x$ to rewrite everything in terms of one function.

5. Using Identities to Solve Equations

If an equation contains mixed trig functions, use identities to reduce to one function.

Example — Solve $2\cos^2 x + \sin x - 1 = 0$ for $0^\circ \leq x \leq 360^\circ$:

Replace $\cos^2 x$ with $1 - \sin^2 x$:
$2(1 - \sin^2 x) + \sin x - 1 = 0$
$2 - 2\sin^2 x + \sin x - 1 = 0$
$-2\sin^2 x + \sin x + 1 = 0$
$2\sin^2 x - \sin x - 1 = 0$
$(2\sin x + 1)(\sin x - 1) = 0$
$\sin x = -\frac{1}{2}$ or $\sin x = 1$
$x = 90^\circ, 210^\circ, 330^\circ$

6. General Solutions

For equations where solutions in all of $\mathbb{R}$ are needed:

$$\sin\theta = k \Rightarrow \theta = \sin^{-1}(k) + 360n^\circ \quad \text{or} \quad \theta = (180^\circ - \sin^{-1}(k)) + 360n^\circ$$

$$\cos\theta = k \Rightarrow \theta = \pm\cos^{-1}(k) + 360n^\circ$$

$$\tan\theta = k \Rightarrow \theta = \tan^{-1}(k) + 180n^\circ$$

where $n \in \mathbb{Z}$ (any integer).

7. Introduction to R sin(x + α) Form

Expressions of the form $a\sin x + b\cos x$ can be written as $R\sin(x+\alpha)$ where:

$$R = \sqrt{a^2+b^2}, \qquad \tan\alpha = \frac{b}{a}$$

This is particularly useful for finding the maximum/minimum of such expressions and solving equations. The full method is developed in P16 (Further Trigonometry).

Example — Express $3\sin x + 4\cos x$ in the form $R\sin(x+\alpha)$:

$R = \sqrt{3^2+4^2} = \sqrt{25} = 5$
$\tan\alpha = \frac{4}{3} \Rightarrow \alpha = 53.13^\circ$
So $3\sin x + 4\cos x \equiv 5\sin(x + 53.13^\circ)$
Maximum value is 5, minimum value is −5.

Exam Tip — Common Exact Values:

θ	$\sin\theta$	$\cos\theta$	$\tan\theta$
0°	0	1	0
30°	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{3}}$
45°	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$	1
60°	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$
90°	1	0	undefined

Key Exam Points for P7:

- Always state the number of solutions expected and check your answers lie in the given interval
- When working in radians, keep answers as exact multiples of π unless told otherwise
- Disguised quadratics — let $u = \sin x$ or $u = \cos x$, factorise, then solve each case
- If $|\sin x| > 1$ or $|\cos x| > 1$, there are no solutions — state this clearly
- Proofs: never cross-multiply across $\equiv$; work on one side at a time

© RAG Learning · Not for redistribution